

Adaptive Multi-Modal Satellite Image Super- Resolution Using Hybrid Deep Learning

B.Tech Project Presentation

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Problem Statement

Most open-source remote sensing data have coarse spatial resolution, and it is advantageous to use different sensors for effective enhancement. However, sensors have differences in spatial resolution, spectral characteristics, noise levels, and often lack spatial or temporal overlap. An adaptive approach is needed to unify these differences.

Objective

-  **To develop** a multisensor super-resolution approach that effectively addresses characteristic differences of different sensors.
-  **To design** a method that combines multiple sensors to improve image resolution, while properly handling differences in spatial resolution, spectral properties, noise, and acquisition conditions.

What is Super Resolution?

Recovery Targets

Recover lost image details, textures, sharp edges, and vital structural information.

Main Objectives

1. Improve Image Clarity
2. Recover Image Details
3. Enhance Texture and Edges

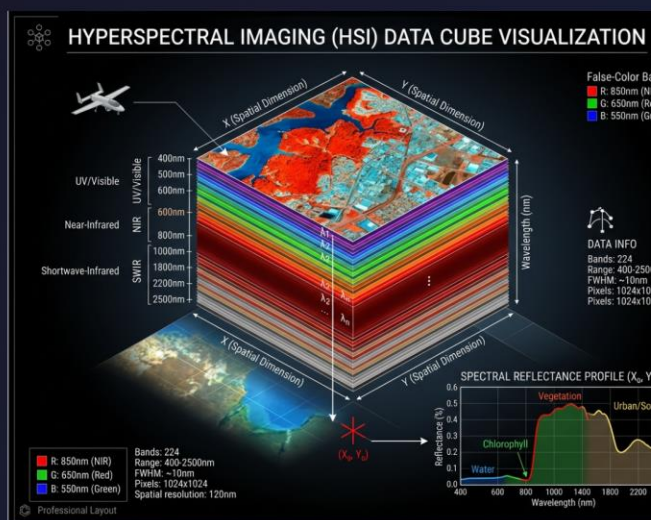
Basic Workflow

LR Image → Feature Extraction → SR Model → Image Reconstruction → HR Image



Hyperspectral Imaging (HSI)

- **RGB:** 3 bands | **HSI:** 100-200+ bands (narrow wavelength ranges)
- Each pixel has a full spectral signature (like a fingerprint)
- **Applications:** Agriculture, Mining, Environmental monitoring
- **Challenge:** HSI has rich spectral info but **LOW spatial resolution**



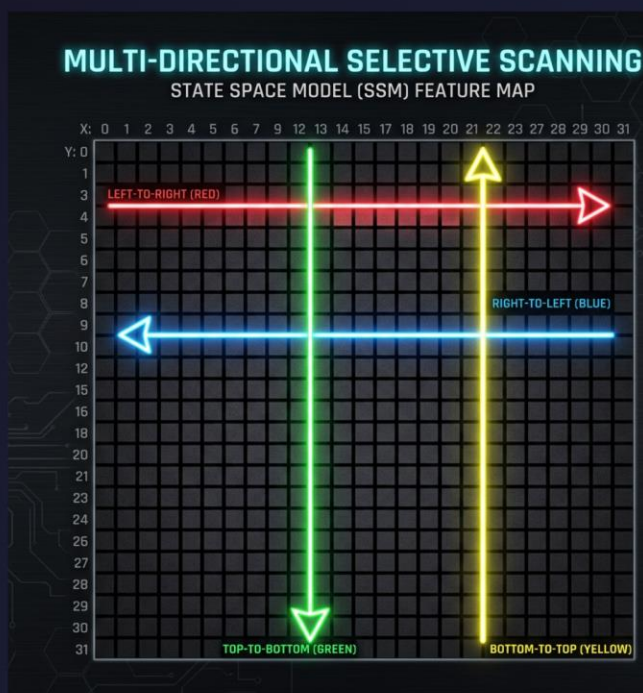
Why Multi-Sensor Fusion?

- Different sensors capture different information about the same scene.
- Each sensor has strengths and weaknesses (e.g., one has good spatial detail, another has rich spectral info).
- Fusing data from multiple sensors combines their strengths.
- **Challenge:** Sensors differ in resolution, noise, spectral range — requiring an **adaptive approach**



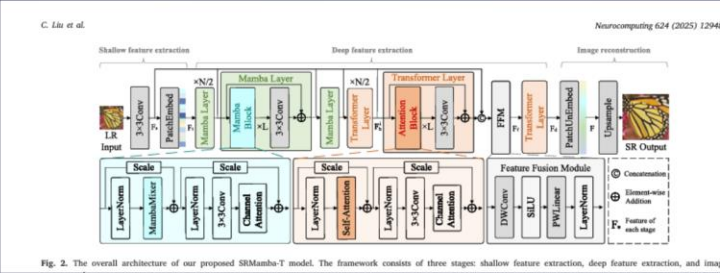
MDSSM — Solving the 2D Scanning Problem

- Mamba natively processes **1D sequences**, but images are **2D**.
- **Solution:** Scan in **4 directions** to capture all spatial relationships.
- Split channels into 4 groups, one per scan direction (saves compute).



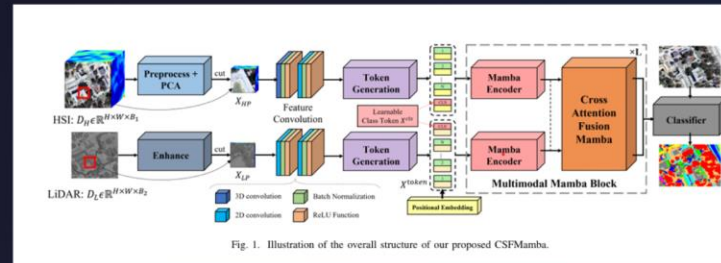
SRMamba-T Architecture (Base Paper)

- **Paper:** "SRMamba-T: Hybrid Mamba-Transformer for SISR" (Neurocomputing 2025)
- **3-Stage:** Shallow Extract → Deep Extract (U-Net) → Reconstruct
- **Key Idea:** Mamba layers early (O(N) efficiency) + Transformer late (global refinement)
- **Our base** — we extend this to



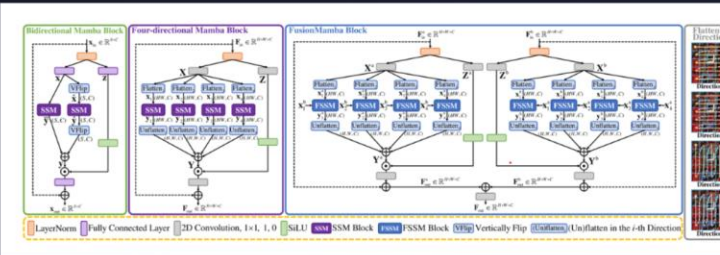
CSFMamba: Cross-State Fusion Mamba

- **Task:** Multi-modal data classification (e.g., HSI + LiDAR)
- **Key Idea:** One sensor controls SSM parameters (B, C) to guide processing of the other.
- **Result:** MUUFL 91.13% OA, Houston2018 93.38% OA



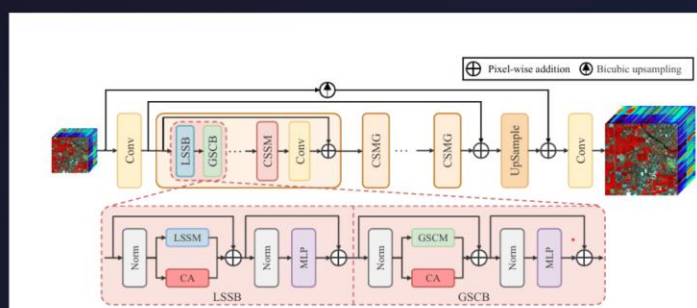
FusionMamba: Dual-Input State Space Fusion

- **Task:** Remote Sensing Image Fusion
- **Key Idea:** FSSM block — one sensor controls, another gets processed.
- Two U-shaped branches (Spatial + Spectral) + Combination branch.
- **Limitation:** Late fusion only —



HSRMamba: Spectral-Aware Mamba for HSI-SR

- **Task:** Single HSI Super-Resolution
- **Key Idea 1:** LSSP — Local 3D window partitioning (prevents pixel forgetting).
- **Key Idea 2:** GSRM — Reorder pixels by spectral similarity before scanning.
- **Limitation:** HSI only — no multi-



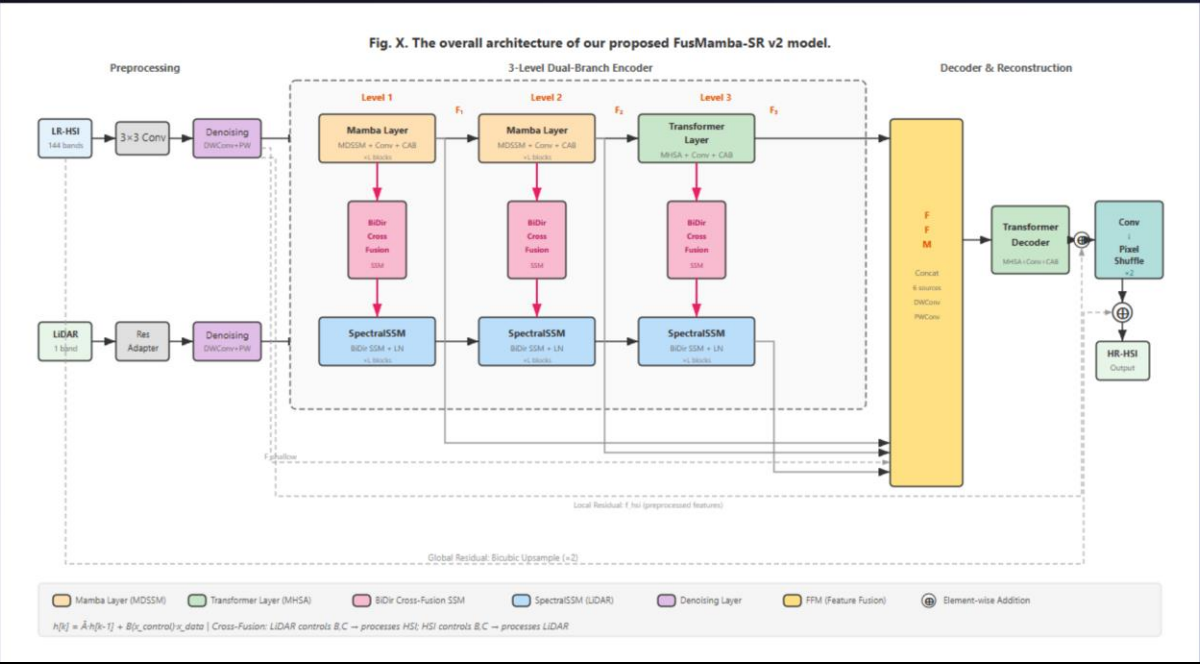
Part II: Our Model — FusMamba-SR v2

FusMamba-SR v2 — Our Novel Architecture

6 Key Contributions:

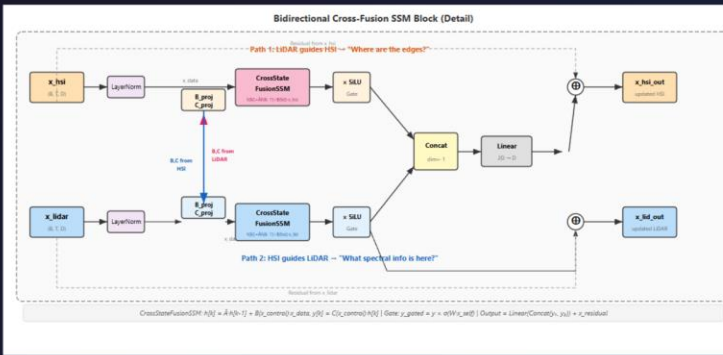
1. **Bidirectional Cross-Fusion SSM** (HSI $\leftrightarrow$ LiDAR)
2. **Multi-Level Fusion** at every encoder stage
3. **Resolution Adapter** for different sensor resolutions
4. **Denoising Layer** with learnable thresholds
5. **QuadLoss** (MSE + FFT + SAM + SSIM)
6. First hybrid Mamba-Transformer for multi-sensor HSI-SR

FusMamba-SR v2 — Complete Architecture



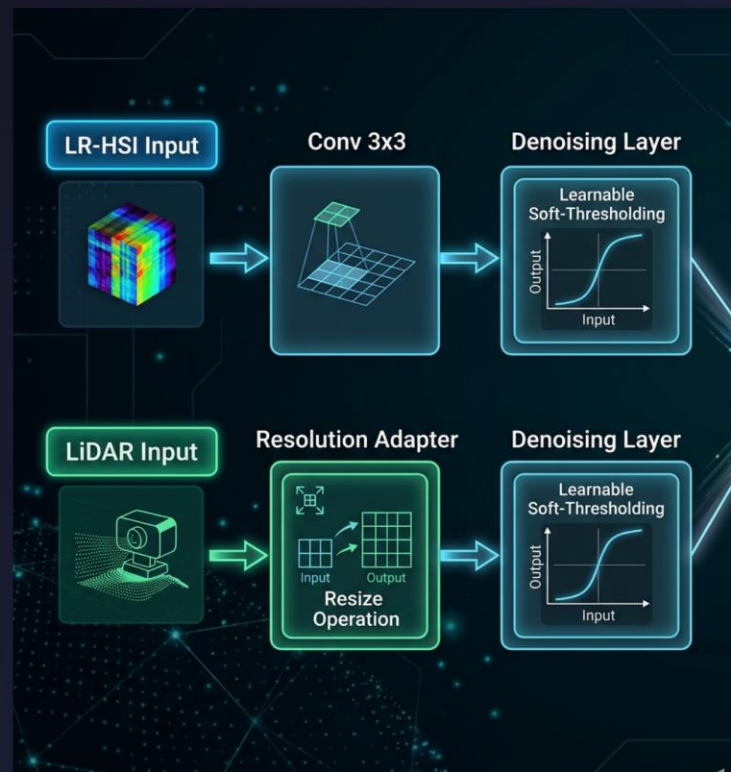
Bidirectional Cross-Fusion SSM (Our Key Innovation)

- **Path 1:** LiDAR guides HSI — "Where are the edges?"
- **Path 2:** HSI guides LiDAR — "What spectral info is here?"
- Gating + Concat + Linear projection
- Applied at **ALL 3 encoder levels** (not just at the end)



Handling Sensor Differences

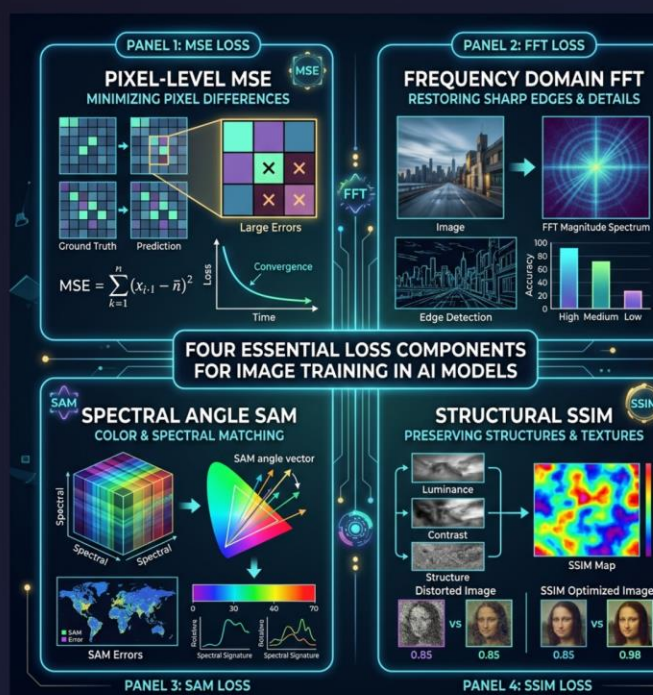
- **Resolution Adapter:** Bilinear resize + Conv (handles different spatial resolutions between sensors)
- **Denoising Layer:** Learnable soft-thresholding per channel



QuadLoss — 4-Component Training Loss

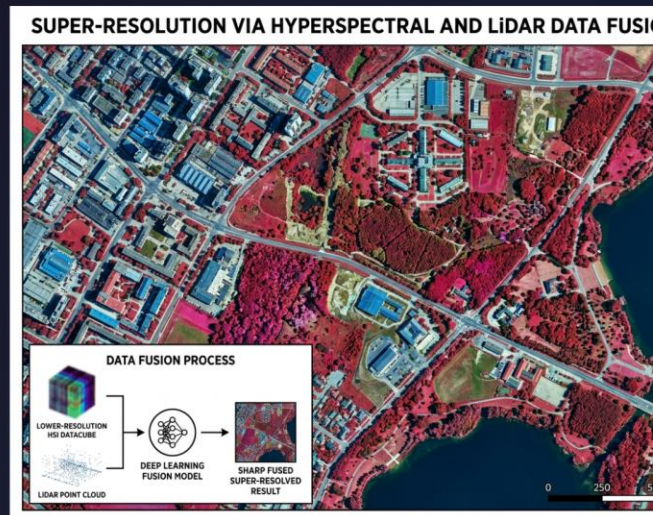
$$L_{\text{total}} = L_{\text{MSE}} + 0.1 \cdot L_{\text{FFT}} + 0.01 \cdot L_{\text{SAM}} + 0.05 \cdot L_{\text{SSIM}}$$

- **MSE**: Pixel accuracy
- **FFT**: Preserves edges and textures
- **SAM (Spectral Angle Mapper)**: Preserves spectral shape (critical for HSI)
- **SSIM**: Structural similarity



Dataset: Houston 2013

- **Sensors:** ITRES CASI (HSI) + Optech Gemini (LiDAR)
- **Data:** HSI: 144 spectral bands, LiDAR: 1 band
- **Size:** 349×1905 pixels, 2.5m GSD
- **Training:** 2360 patches (32×32)



Training Setup

Parameter	Value
Model Params	564,992
Optimizer	Adam ($\beta_1=0.9$, $\beta_2=0.99$)
Learning Rate	2×10^{-4} with MultiStepLR
Batch Size	8
Patch Size	32×32
Scale	$\times 2$
Hardware	NVIDIA T4
AMP	Mixed Precision (FP16)

Experimental Results

Model	Task	Params	Key Metric
SRMamba-T	Single Image SR	653K	33.20 dB (Urban100)
FusionMamba	RS Image Fusion	~1M	35.63 dB (Pavia)
HSRMamba	HSI-SR	~2M	40.28 dB (Chikusei)
FusMamba-SR v2	Multi-Sensor HSI-SR	564K	36.91 dB (Houston)

Training results visualization — to be added after full training

Summary & Future Work

Contributions: Lightweight (564K params), Novel Cross-Fusion SSM, robust Multi-Level Fusion, Multi-Sensor Preprocessing, and QuadLoss.

Future Work:

- Extend to $\times 4$ and $\times 8$ scales
- Real-world degradation model (noise, blur, spectral response)
- CUDA-optimized selective scan for $10\times$ speedup
- Test on more datasets (Chikusei, WDC, Botswana)

Thank You!

Questions?

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